

RapidBoxForest: Link-Envelope-Guided Box Forests for Fast Multi-Query Manipulation

TIAN, Yu and Ren, Hongliang

Abstract—Fast repeated-query manipulation benefits from reusable volumetric planning structure, but high-quality IRIS-style convex regions are expensive to construct and rebuild. This paper presents RapidBoxForest (RBF), a link-envelope-guided box-forest planner for fast multi-query manipulation. The central geometric primitive maps endpoint interval envelopes to link interval envelopes via capsule Minkowski decomposition and convex-hull generator bounding applied to *C-space box validation and multi-query reuse*: endpoint interval envelopes bound rounded-link endpoint motion over a joint box, and a radius-lifted link envelope provides an inexpensive filter for box validation and collision rejection. A rapidly-exploring random tree (RRT)-inspired grower builds a scene-specific box forest into a queryable corridor graph, and a Lifelong Envelope Cache Tree (LECT) selectively caches kinematics-keyed envelope records when replay is cheaper than recomputation. In the regenerated endpoint-envelope experiment, each source is evaluated on 6000 shared IIWA14 joint boxes, with 1000 boxes at each of six fixed widths. The public IFK spelling, which now subsumes the former AAFK alias, averages 4.53 μ s per endpoint box with no observed reference under-coverage. Hierarchical IFK tightens the mean endpoint volume by up to 76.0% relative to IFK at 72.2 μ s. CritSample and MC give smaller boxes but are non-certified and show observed under-coverage against the finite reference union. These results bound the quantitative claims in this version to the endpoint-envelope primitive; full planner-level cache, dynamic-update, and cross-planner tables are not carried forward here.

Index Terms—Motion planning, multi-query planning, link interval envelopes, box-region planning, final audit.

I. Introduction

Motion planning for high-degree-of-freedom (DOF) manipulators becomes especially demanding when a system must answer many related queries rather than one isolated instance. In warehouses, packing cells, and shared workcells, robot kinematics stay fixed while obstacles, task endpoints, or local scene edits change. The central question is how to construct reusable *C-space* planning structure quickly enough that repeated queries can benefit from region-level search without paying a large scene-specific preprocessing cost.

Iterative Regional Inflation by Semidefinite Programming (IRIS) and Graph-of-Convex-Sets (GCS) methods

show why free-region construction is valuable: once good convex regions are available, graph search or convex optimization can produce high-quality motions over those regions [1], [2], [3]. The limitation is construction cost. In high-dimensional manipulator scenes, inflating and validating high-quality convex regions can dominate the planning budget and must often be repeated when the scene changes. Sampling-based planners such as probabilistic roadmap (PRM) and rapidly-exploring random tree (RRT) methods avoid this region-construction cost [4], [5], [6], [7], but their reusable structures are usually zero-volume graphs, so repeated queries still pay for local collision checking and do not obtain a reusable free-region cover.

RBF targets a different design point by constructing simpler box regions quickly. Instead of inflating general polytopes, it grows axis-aligned boxes in configuration space, connects accepted boxes into a graph, and answers later queries by searching the resulting box corridor. This trades per-region expressivity and sometimes path length for cheap construction, simple adjacency, and direct reuse across many queries. The intended claim is a system-level one: a fast box-region planning pipeline can be a useful final-audited alternative when multi-query latency matters more than obtaining the shortest path from a more expensive region decomposition.

The key technical ingredient is a link interval envelope computed from endpoint interval envelopes. For rounded link models, RBF reduces link motion over a joint box to the motion of a zero-radius centerline plus a final radius lift. Endpoint interval envelopes bound the endpoint trajectories over the joint box; link interval envelopes then enclose the swept rounded link and can be stored as axis-aligned bounding boxes (AABBs), *k*-DOP slabs, or SupportHull records. Strict interval sources provide an optional conservative validation path, while tighter practical sources are used as fast filters and discharged by the same final scene audit used for all reported planners.

On top of this geometric primitive, the grower borrows the rapid frontier-expansion idea of RRT: samples choose useful frontiers, a local region oracle materializes a free box around the projected seed, and accepted boxes extend a connected box forest. The result is also an adaptive cell-decomposition view of manipulator *C-space*, but one biased toward the regions needed by the current repeated-query workload. A Lifelong Envelope Cache Tree

The authors are with the Department of Electronic Engineering, The Chinese University of Hong Kong (CUHK), Hong Kong SAR, China. E-mail: ty1997@link.cuhk.edu.hk.

(LECT) then memoizes scene-independent endpoint and link-envelope evidence keyed by robot kinematics and joint intervals; it is an acceleration layer for repeated materialization, not a separate planning claim.

The evaluation therefore asks how the endpoint-envelope source family behaves under the adopted protocol. The current quantitative axes are endpoint construction time, endpoint AABB volume, and the observed under-coverage gap against a finite reference union on shared IIWA14 joint boxes. Planner-level cache, dynamic-update, cross-planner, and multi-robot generalization studies remain future work for this paper track.

We therefore position the paper as a planning-systems contribution enabled by a geometry primitive. The endpoint-to-link envelope is the main geometric idea; the method shows how that primitive supports a reproducible low-build-cost multi-query design point under final audit.

The paper’s main contribution is a reproducible final-audited multi-query planning system and its measured design point. It is supported by three concrete elements:

- 1) *An endpoint-to-link envelope primitive for C-space box validation.* Capsule Minkowski decomposition and convex-hull generator bounding are individually classical facts already combined in articulated-body CCD work [8], [9], [10]. RBF applies this two-step pipeline to *C-space box growth and multi-query reuse*: a switchable endpoint source family—AA-backed interval forward kinematics (IFK), hierarchical IFK (HIFK), CritSample, analytical extrema—compact per-representation link records (AABB, SupportHull) for staged filtering, and a kinematic-keyed LECT cache for cross-scene envelope reuse. No prior work has been found applying this pipeline to *C-space box validation* or *multi-query planning reuse* (sections III and VI-A).
- 2) *A self-contained link-envelope-guided box-forest planner.* Scene-specific *C-space box forests* are grown through seeded frontier expansion, consolidated into a connected corridor graph, and queried by graph search and local refinement. LECT provides an optional kinematic-keyed cache for repeated envelope materialization across builds (sections IV to VI).
- 3) *An implementation-aligned endpoint-envelope evaluation.* Experiment 1 regenerates the endpoint-source sweep on shared fixed-width IIWA14 joint boxes and reports time, volume, certification status, and finite reference-gap diagnostics from the current implementation. Older planner, cache, dynamic-update, and cross-planner tables are not carried forward (section VI-A).

The paper’s evidence chain is: Section III develops the endpoint and link interval-envelope primitive for fast box validation; section IV describes the cache optimization that reuses expensive envelope records; section V turns link-envelope-filtered boxes into a rapidly expanding box-forest planner; and section VI evaluates the regenerated

endpoint-envelope source trade-off under the adopted protocol.

II. Related Work

A. Convex Region Planning and Graphs of Convex Sets

Convex region planners approach motion planning by constructing a certified cover of free space and then solving graph search or convex optimization on top of that cover. IRIS introduced the now-standard idea of computing large collision-free convex regions by alternating separating hyperplanes and region inflation [1]. Later work extended the same line into configuration space, improved region quality, or improved graph compactness through visibility-graph and clique-cover constructions [11], [12], [13]. Marucci *et al.* made the Graph-of-Convex-Sets formulation particularly influential by showing that motion planning can be cast as optimization over a graph of certified convex regions [2], [3].

RBF is aligned with this family in objective but not in geometric primitive: it uses axis-aligned boxes in *C-space* rather than general polytopes, trading per-region expressivity for lower build overhead and simpler adjacency bookkeeping. The resulting cover is coarser than a high-quality convex decomposition, but it avoids repeated hyperplane-based region inflation and is easier to rebuild under repeated-query workloads. The comparison in this paper therefore concerns fast reusable free-region construction versus tighter, more scene-coupled regional convexification, rather than box and polytope representations in isolation.

B. Swept Volume Approximation and Capsule Geometry

Computing the swept volume of a moving rigid body has been studied extensively in computational geometry and computer-aided manufacturing (CAM) communities [14], [15]. For a convex body $P = \text{conv}(v_1, \dots, v_m)$, bounding each generator trajectory by a conservative outer set $\hat{\Gamma}_i(I)$ yields

$$\mathcal{S}_I(P) \subseteq \text{conv}\left(\bigcup_i \hat{\Gamma}_i(I)\right),$$

a bound that is implicit in oriented bounding box (OBB) tree construction for swept mesh primitives [16].

For capsule links this bound reduces to bounding only the two segment endpoints. The Proximity Query Package (PQP) exploits exactly this Minkowski decomposition ($C = S \oplus B_2(r)$, sweep of C reduces to sweep of S plus radius lift) [17]; GJK and its robust implementation by van den Bergen apply the same identity for distance queries [18], [19]; the Flexible Collision Library (FCL) generalises the swept-sphere interface to a full collision library [20].

C. Interval Forward Kinematics and the CCD Endpoint-Bounding Pipeline

Interval arithmetic [21], [22] provides inclusion-monotone enclosures of real-valued functions over

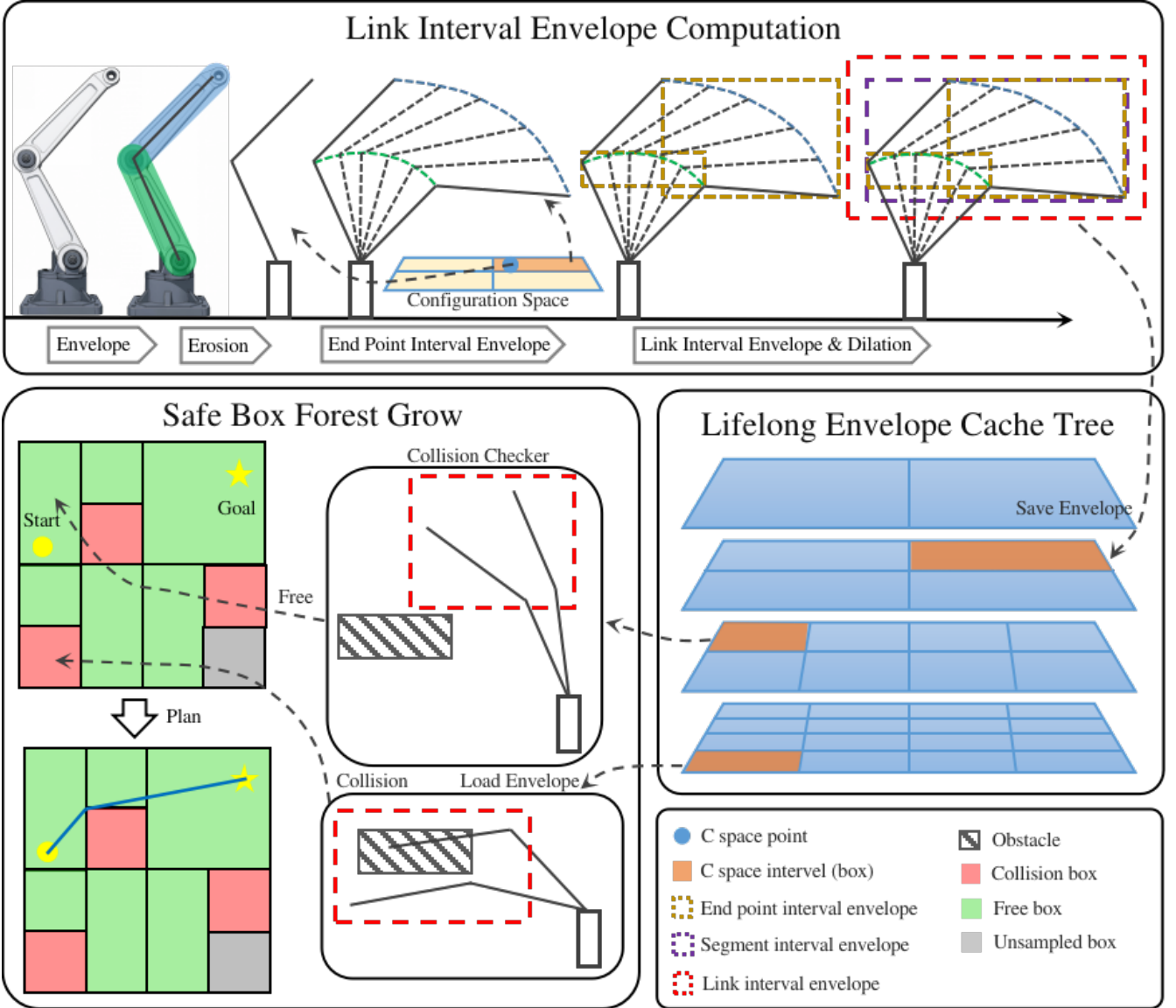


Fig. 1. RBF pipeline for link-envelope-guided multi-query planning. Top: endpoint interval envelopes induce link interval envelopes over a C -space box. Middle: LECT optionally reuses kinematics-keyed endpoint and link-envelope records across repeated builds. Bottom: scene filtering and an RRT-inspired grower expand a connected box-region forest for lightweight corridor queries and final audit.

parameter intervals. Snyder applied interval arithmetic to parametric geometry to compute conservative bounding boxes for curves and surfaces without mesh sampling [23]. Merlet extended DH-chain interval propagation to certified workspace analysis for parallel and serial manipulators [24], [25].

These ideas meet directly in the continuous collision detection (CCD) literature for articulated bodies. Redon, Kim, Lin, and Manocha [8] model each robot limb as a line-swept sphere (LSS, i.e. capsule) and propagate interval Denavit-Hartenberg (DH) transforms along the kinematic chain to obtain *interval vectors bounding the two LSS endpoint positions* over the motion time step; the swept capsule is then bounded by padding the endpoint bounding box with the link radius. This is explicitly the two-step pipeline

$$\text{interval FK endpoint bounding} \rightarrow$$

$$\text{bbox}([\mathbf{p}_i] \cup [\mathbf{p}_{i+1}]) \oplus B_2(r) = \text{link swept-capsule bound}$$

that is the geometric core of RBF. The journal extension by Redon, Lin, Manocha, and Kim [9] applies the same framework to general robot arms and kinematic trees; that work has accumulated over 240 citations and is a standard CCD reference. Zhang *et al.* [10] replace interval arithmetic with Taylor models in the same pipeline to reduce wrapping errors, analogous to how RBF’s CritSample and analytical sources tighten the IFK bound. Schwarzer *et al.* [26] address a multi-query motivation similar to RBF’s—reusing collision evidence across planning calls—but at the path-segment level rather than as volumetric C -space records. Corrales *et al.* [27] apply sphere-swept line (SSL) bounding volumes dynamically for human-robot safety monitoring using the same structural observation that bounding a capsule’s motion reduces to bounding its two endpoints.

D. Novelty Boundary: Prior CCD vs. RBF

The two-step endpoint-interval-to-swept-capsule pipeline is therefore unambiguously present in the CCD literature for time-step motion. The differences that motivate the RBF contribution claim are: (a) the parameter domain is a *C-space joint-box interval* rather than a trajectory time step, so the goal is to classify whether *any* configuration in the box is collision-free, not whether a single given path collides; (b) the endpoint source is a switchable interface—IFK (conservative), CritSample (practical), and analytical extrema (tightest)—rather than a fixed algorithm, allowing the conservative vs. performance trade-off to be controlled without changing the link-representation or planner layers; (c) compact per-representation link records (AABB, SupportHull) are organised for staged filtering during box growth, a knob absent from CCD libraries; and (d) the endpoint and link envelope records are organised in LECT, a persistent *kinematic-keyed* cache that reuses evidence across multi-query builds in related scenes, a reuse level not addressed by CCD libraries or path-segment caches. No prior work has been found that applies the two-step endpoint bounding pipeline to *C-space* box growth or to multi-query volumetric planning reuse. RBF is complementary to the CCD stack: link envelopes reduce repeated geometric work on candidate boxes, while discrete scene checks remain the final authority for the reported paths.

E. Multi-Query Reuse and Adaptive Configuration-Space Decomposition

Multi-query planners typically amortize effort by reusing a graph structure such as a roadmap. PRM is the classical example: build a connectivity graph once and reuse it for many start-goal pairs [6], [28]. LazyPRM, lazy edge checking, and certificate-based planners postpone or cache edge validation so that online search does not repeatedly evaluate the same discrete local connections [29], [30]. Experience-based and trajectory-library systems reuse prior solutions or local motion snippets, often as initialization or bias for a new query rather than as a conservative decomposition of a region of configuration space [31].

RBF operates at a different reuse level. It grows and reuses volumetric boxes rather than only paths, binary collision outcomes, or zero-volume edges. PRM-like methods reuse connectivity and must revalidate affected edges after scene changes. Convex-region planners reuse higher-quality regions, but those regions are usually coupled to the obstacle layout in which they were constructed.

The method also inherits the central idea of adaptive cell decomposition and subdivision planning: represent free space by cells, split cells where the current evidence is insufficient, and search the adjacency graph induced by the accepted cells [32], [33]. The difference is the operating regime. Classical cell-decomposition examples are often low-dimensional workspace or explicit *C-space*

constructions, whereas RBF applies the same adaptive-cell idea to high-dimensional manipulator *C-space* through a link-envelope oracle. Scene-specific obstacle filtering is layered on top of reusable kinematic envelope evidence, with LECT available only as a cache for expensive records. The trade-off is explicit: RBF accepts axis-aligned boxes and over-approximation in exchange for simple validation, adjacency maintenance, and repeated-query throughput.

F. Sampling-Based Planning and Trajectory Optimization

Sampling-based planners remain standard baselines when the goal is to solve a single query quickly. RRT-Connect is particularly effective in manipulator planning because bidirectional growth often succeeds with very little global structure [34]. Asymptotically optimal and informed variants such as RRT*, Batch Informed Trees (BIT*), and Fast Marching Tree (FMT*) improve path quality and search efficiency when more time is available [7], [35], [36], [37]. The Open Motion Planning Library (OMPL) has made these methods easy to deploy and benchmark in a common software stack [38], [39]. However, standard RRT variants produce single-use zero-volume evidence and typically restart from scratch on the next query.

By contrast, RBF uses rapidly exploring, adaptive expansion to sample and materialize *C-space* boxes from the envelope oracle, producing a volumetric corridor structure that can be queried repeatedly within the same environment. Trajectory optimizers address a different stage of the pipeline. CHOMP, STOMP, TrajOpt, and Gaussian-process motion planners refine an initial path by local descent or sequential convex optimization [40], [41], [42], [43]. RBF uses these methods as complementary points of comparison: its aim is to provide a reusable volumetric backbone on which graph search and local post-processing can operate.

III. Link Interval Envelopes

This section develops the main geometric primitive used by RBF: a map from a manipulator joint box to compact link interval envelopes. The goal is to make box validation cheap enough for repeated-query planning. A strict interval source gives a conservative validation option, while tighter sampled or support-based variants are practical filters whose reported paths are discharged by the common final audit. Thus the strongest free-region statement in this section is conditional on the conservative source and on paths remaining inside the validated box union; the main planning experiments report final-audited performance separately. The construction first bounds endpoint trajectories and then applies a constant-radius lift for rounded link geometry.

Here $\mathcal{Q}$ denotes the joint-limit box explored by RBF, $I \subseteq \mathcal{Q}$ one joint interval, and $\mathcal{O} \subset \mathbb{R}^3$ the obstacle set. Endpoint envelope $E_k(I)$ outer-bounds endpoint k , and link envelope $B_\ell(I)$ outer-bounds the swept volume of link ℓ over the same interval. Together they define the

reusable planning primitive $I \mapsto \{E_k(I)\} \mapsto B_\ell(I)$. When the endpoint source is itself conservative, the resulting link envelope is conservative as well; when a tighter empirical source is used, the same representation serves as a fast broad- or mid-phase filter before final scene auditing rather than as a standalone certificate.

A. Geometric Foundation

Let $T(q)x = R(q)x + t(q)$ denote the rigid transform induced by a configuration $q \in I$. For a reference body $C_0 \subset \mathbb{R}^3$, its pose at q is $C(q) = T(q)C_0$. We write $S_I(C_0)$ for the exact swept workspace set over I and $C_I(C_0)$ for its convex envelope. RBF usually works with the convex envelope because the downstream representations are AABBs, hulls, or fixed-direction slab bounds. The following construction combines two classical computational-geometry facts—capsule Minkowski decomposition and convex-hull generator bounding—in a new planning context. Both facts are used individually in swept-sphere collision libraries and in CCD work for articulated bodies [8], [9]; the novelty here is applying their combination to *C-space box validation and multi-query reuse* rather than to single-query time-step CCD.

Minkowski sum separation (radius isolation). For a capsule $C = S \oplus B_2(r)$, rigid motion commutes with the fixed Euclidean ball, giving exact two-stage sweep bounds:

$$S_I(C) = S_I(S) \oplus B_2(r), \quad C_I(C) = C_I(S) \oplus B_2(r).$$

RBF therefore bounds the zero-radius seed first and applies the link radius at the end. This identity is exact for capsules; for sharp polytopes it requires the shape to be a rounded offset body.

Convex-hull generator bound. For a convex body $P = \text{conv}(v_1, \dots, v_m)$, bounding each vertex trajectory by an outer set $\hat{\Gamma}_i(I)$ suffices to bound the swept polytope:

$$S_I(P) \subseteq \text{conv}\left(\bigcup_i \hat{\Gamma}_i(I)\right).$$

For a capsule link the seed is the center segment $[a, b]$, so only two endpoint trajectories need to be bounded.

Combining these two facts yields the practical recipe. For a capsule link $[a, b] \oplus B_2(r)$, RBF bounds only the two endpoint trajectories, forms their convex hull, and applies the radius lift:

$$\hat{C}_I([a, b] \oplus B_2(r)) = \text{conv}(\hat{\Gamma}_a(I) \cup \hat{\Gamma}_b(I)) \oplus B_2(r).$$

The numerical task reduces to computing reusable envelopes for two endpoint trajectories per link, fully determined by endpoint-trajectory enclosures plus the chosen downstream representation.

Proposition 1 (Endpoint envelopes induce link interval envelopes). *Consider a rounded link $L = [a, b] \oplus B_2(r)$ and a joint interval I . If endpoint envelopes $E_a(I)$ and $E_b(I)$ satisfy $\Gamma_a(I) \subseteq E_a(I)$ and $\Gamma_b(I) \subseteq E_b(I)$, then*

$$S_I(L) \subseteq \text{conv}(E_a(I) \cup E_b(I)) \oplus B_2(r).$$

Consequently, any outer representation of the right-hand side is a valid link interval envelope for the whole joint box I .

Proof. For each $q \in I$, the center segment of the moved link is the convex hull of the two moved endpoints, so $T(q)[a, b] \subseteq \text{conv}(\Gamma_a(I) \cup \Gamma_b(I))$ over the entire interval. The endpoint inclusions place this set inside $\text{conv}(E_a(I) \cup E_b(I))$. Adding the fixed radius ball commutes with the rigid-motion sweep of the capsule, giving the stated containment. $\square$

B. Endpoint Interval Envelopes

Figure 1 illustrates the AABB extraction pipeline. Endpoint interval axis-aligned bounding boxes (iAABBs) are computed by interval forward kinematics (IFK) or tighter certified AA-based variants, combined into a conservative zero-radius centerline envelope, and padded by the link radius before collision querying. The remaining numerical task is to outer-bound each endpoint trajectory over I by a reusable set $E_k(I)$. Endpoint envelopes are the common interface between kinematic propagation and the downstream link-envelope representations.

For endpoint k , let $\Gamma_k(I)$ denote its exact workspace trajectory over interval I . Any conservative set that contains $\Gamma_k(I)$ is an endpoint interval envelope. In the baseline implementation this envelope is simply an axis-aligned bounding box, because boxes are cheap to store, easy to combine, and directly reusable by all downstream link representations.

a) *Interval forward kinematics (IFK).*: In the public API used in this paper, `EndpointSource.IFK` is the spelling for the IFK_AA-backed certified endpoint source. It propagates affine-arithmetic workspace bounds along the DH chain and remains the cheapest certified endpoint source in the active paper track, but it can loosen on wide boxes or long chains because one evaluation still accumulates residual over-approximation across the full joint box.

b) *Hierarchical IFK (HIFK).*: HIFK further reduces wrapping by recursively bisecting the joint box and evaluating AA-FK at the leaves before hulling the child endpoint boxes. This is not the same as the unsplit IFK baseline: a true depth-1 HIFK evaluation already splits once and hulls two leaves, whereas IFK performs one unsplit AA-FK pass on the original box. The mechanism study in section VI-A reports two representative split depths, HIFK_3 and HIFK_5, to quantify how additional hierarchical depth trades runtime for tighter certified endpoint boxes.

c) *Incremental FK state reuse.*: Tree descent also carries a transient FK state: interval-valued prefix transforms $[T^{0:0}], \dots, [T^{0:n}]$ and the per-joint DH interval matrices used to build them. The full prefix chain is computed once at the root or after a restart; when only I_k changes, prefixes before k remain valid and only the suffix is recomputed. This short-lived state is not the persistent cache itself; it only keeps endpoint materialization cheap before LECT decides whether to retain the evidence.

d) *Critical-sample envelopes.*: The second source estimates the endpoint box from carefully chosen boundary

and interior samples. In practice these samples include interval endpoints, turning-point candidates near multiples of $\pi/2$, and a few midpoint or near-boundary probes. This often captures extrema that IFK wraps away, at modest extra cost. The limitation is completeness: separable per-joint samples do not certify coupled extrema. CritSample is therefore a performance-oriented source whose reported planning paths are discharged by the final scene audit unless an external certificate is added.

e) Analytical extrema.: The third source solves endpoint-coordinate extrema from the trigonometric structure of the chain. Because each revolute-joint transform is affine in $(\cos q_j, \sin q_j)$, an endpoint coordinate becomes a simple sinusoid when one joint is free and a low-degree bilinear trigonometric polynomial when two joints remain free. This yields a structured solver: vertices and 1D boundary edges can be enumerated exactly, 2D faces reduced to polynomial root finding, and higher-dimensional residual cases handled by local descent with interval verification or branch-and-bound closure. Analytical bounds are the tightest of the three sources, but also the most expensive, so they are better suited to offline certification and profiling than to the online path.

Taken together, these sources define a clear spectrum: IFK is the cheapest certified AA baseline, HIFK_3 and HIFK_5 offer progressively tighter certified outer bounds at increasing cost, CritSample is the cheapest tighter advisory bound, and the analytical source is the tightest but most expensive. Endpoint envelopes are the right intermediate abstraction: they are the state variable of the joint-box-to-envelope map, so the downstream link-envelope layer stays fixed while the upstream source changes with the certification and runtime budget.

C. Link Interval Envelopes

Once endpoint envelopes are available, the remaining design question is how to represent the swept link volume compactly for box growth, collision filtering, and optional reuse. For a capsule link, RBF does not reason about the thick link directly. It first bounds the zero-radius center segment between the two endpoint envelopes and then pads that centerline envelope by the link radius. A conservative link interval envelope $B_\ell(I)$ is any set that contains the full swept capsule over interval I . In the default box-based implementation this is written as $B_\ell(I) = B_\ell^\circ(I) \oplus \mathcal{R}(r_\ell)$, where $B_\ell^\circ(I)$ bounds the centerline sweep and the padding operator $\mathcal{R}$ becomes B_∞ for ABB envelopes.

The following representations share the same endpoint evidence but preserve different amounts of downstream geometric detail.

a) ABB envelope.: ABB envelopes are the simplest option. The ordinary LinkIAABB construction takes the two endpoint envelopes, wraps them in one box, and pads that box by the link radius:

$$B_\ell(I) = \text{bbox}(E_{\ell-1}(I) \cup E_\ell(I)) \oplus B_\infty(r_\ell).$$

This representation can be tightened by subdividing the reference center segment into S pieces, bounding each piece separately, and retaining the union of the padded sub-envelopes. Larger S preserves more directional variation along the link while keeping reconstruction cheap once endpoint envelopes are available. The main caveat is that the subdivision must remain explicit: if all sub-envelopes are collapsed back into one global ABB, much of the gain disappears.

b) SupportHull envelope.: When the endpoint envelopes are convex, their convex hull preserves directional structure that a single ABB discards. SupportHull is the compact runtime realization of that idea: instead of storing an explicit dense hull, it keeps the two endpoint ABBs and the link radius as a support-function record. For a query direction d , the support point is the better of the proximal and distal ABB support points, and collision refinement against an obstacle ABB uses a fixed-size Gilbert–Johnson–Keerthi (GJK) narrow phase with the obstacle inflated by the link radius and safety margin. This retains much of the convex-hull directional fidelity while keeping the default record compact.

In this way, the link-envelope layer is separate from the upstream kinematics solver. Sections III-A to III-C define a modular endpoint-to-link envelope map from joint-space boxes to reusable workspace filters. Once endpoint envelopes are available, the remaining design choice is the representation and the corresponding planning-time cost/tightness trade-off.

IV. Lifelong Envelope Cache Tree

The planner can materialize the envelopes of section III online. LECT is a secondary optimization for workloads in which repeated builds revisit similar joint boxes or use envelope representations expensive enough to replay. It stores scene-independent endpoint/link-envelope evidence keyed by robot kinematics and joint intervals. Scene-specific obstacle tests, box labels, graph connectivity, and final-audit outcomes remain outside the cache. This section therefore records only the cache interface used by the planner; the scientific claim of the paper does not depend on LECT providing broad cross-scene transfer.

A. Cache Key and Split Policy

LECT is a dynamically grown binary KD tree over the joint-limit box $\mathcal{Q}$. Each node represents one joint interval and stores interval bounds, split metadata, cached endpoint evidence from which link envelopes can be reconstructed, and residency state used by the implementation. The useful property is that the stored evidence is *kinematics-keyed* rather than *scene-keyed*: later scenes may replay a materialized envelope record, but they must rerun their own scene-local obstacle tests. The current storage format identifies nodes by 64-bit breadth-first heap ids (0 at the root and $2p + 1, 2p + 2$ for children). To keep this structural id scheme safe, the implementation

enforces a configurable maximum tree depth $D_{\max} \leq 64$, with default $D_{\max} = 64$.

The split policy is envelope-aware rather than purely geometric. For a parent interval I , let I_d^- and I_d^+ be the two children obtained by splitting joint dimension d at its midpoint, and let $\mathcal{V}(I) := \sum_{\ell} \text{vol}(B_{\ell}(I))$ be the cumulative downstream envelope volume. The default best-tighten score chooses the split with the smallest worst-child envelope burden,

$$d^* = \arg \min_d \max \left\{ \sum_{\ell} \text{vol}(B_{\ell}(I_d^-)), \sum_{\ell} \text{vol}(B_{\ell}(I_d^+)) \right\}.$$

The implementation uses this score in a depth-synchronous schedule so that all nodes at the same depth share one split decision. Candidate dimensions are also filtered by FK reachability: splitting a joint beyond the last active frame that contributes to any link endpoint cannot tighten the endpoint envelopes and is skipped.

B. Runtime Use and Validation Boundary

LECT is queried by FINDFREEBOX, not by the graph search itself. Given a joint interval, the cache either replays endpoint evidence or materializes it online, then derives the requested link representation. Directional records are used as refinement evidence rather than as standalone free-space certificates: the runtime first evaluates padded link AABBs, then uses SupportHull/GJK only for survivors of the AABB test. When a node is split, child envelopes may refine the parent witness, but any compact parent record remains representation-dependent and is rechecked against the current obstacle set before a box can enter the forest.

C. Storage and Warm-Build Semantics

Persistence is selective. Endpoint evidence is the common reusable record, but simple IFK+AABB evidence can be cheaper to recompute online than to reload. SupportHull records are stored only when their tighter filters are expected to amortize storage and lookup cost. Implementation details such as node arrays and residency management are engineering choices at the implementation level; the paper does not claim runtime benefit from sector symmetry, because the cache mechanism operates in canonical-identity mode with an empty symmetry descriptor.

The prewarm phase follows the same separation. It materializes FK/link-envelope evidence only on the target leaf layer and reconstructs upper layers by child-hull propagation during checkpoint. This avoids charging direct FK at multiple depths for the same cache and keeps parent witnesses tied to the leaf records that generated them. Exact leaf evidence and propagated child-hull evidence are both treated as envelope records by the replay path; provenance does not create a separate reuse or validation mode.

In the warm-build path, LECT reuses only kinematic envelope evidence. On a cache hit, the current link-envelope representation is reconstructed from stored endpoint records and then revalidated against the current

obstacle set. A warm build still regrows and revalidates scene-specific RBF boxes and never treats a persisted box label as a free-space certificate. In section V, LECT serves only as the materialization layer; obstacle checking, occupancy metadata, adjacency bookkeeping, and regrowth reactions to changing $\mathcal{O}$ remain inside the box-region forest routines. For the reviewer-sensitive Shelf+IIWA protocol, the external d18 cache is treated as read-only during evaluation: warm shelf rows disable active-database backfill and therefore cannot silently accumulate new online envelope evidence while being measured. The balanced random-scene study keeps that writeback path enabled because it measures cache growth on novel scenes rather than a fixed held-out shelf target.

V. RapidBoxForest Construction

This section describes the planning layer of RBF. The algorithm grows a scene-specific forest of C -space boxes and exports the forest as a graph for repeated corridor queries. The growth loop is RRT-like, but the state added to the data structure is a free region rather than a point or edge: samples select frontiers, the link-envelope oracle materializes a usable box, and accepted boxes extend a connected region graph. In adaptive cell-decomposition language, the boxes are cells, FindFreeBox is the local refinement and classification primitive, and the forest graph is the adjacency structure used for query search. The link interval envelopes of section III and the optional LECT cache of section IV serve this planner as collision-filtering and materialization tools.

Throughout this section, a validated box means a box accepted by the selected scene-validation policy. In strict conservative mode, paths that remain inside the validated box union inherit the corridor statement below. In all reported planning experiments, paths that leave this union through repair, smoothing, or external composition are charged where used and passed through the same final scene audit.

Conceptually, the build process has three stages. First, a seed-guided local oracle adaptively refines and classifies a C -space cell around a selected configuration. Second, repeated frontier expansion grows these cells into a connected forest, giving the method its region-valued RRT/PRM character. Third, consolidation coarsens redundant boxes before the structure is exported as a query graph. The consolidation pass is part of the reusable build phase and its time is included in reported build measurements.

A. Construction Objective

Let $\mathcal{Q}$ denote the joint-limit box and let $\mathcal{O}$ denote the current obstacle set. The construction objective is to build a finite family

$$\mathcal{F} = \{B_1, \dots, B_N\}, \quad B_i \subseteq \mathcal{Q},$$

such that every B_i is a validated free-space box, the union $\bigcup_i B_i$ covers as much useful query-relevant free space as

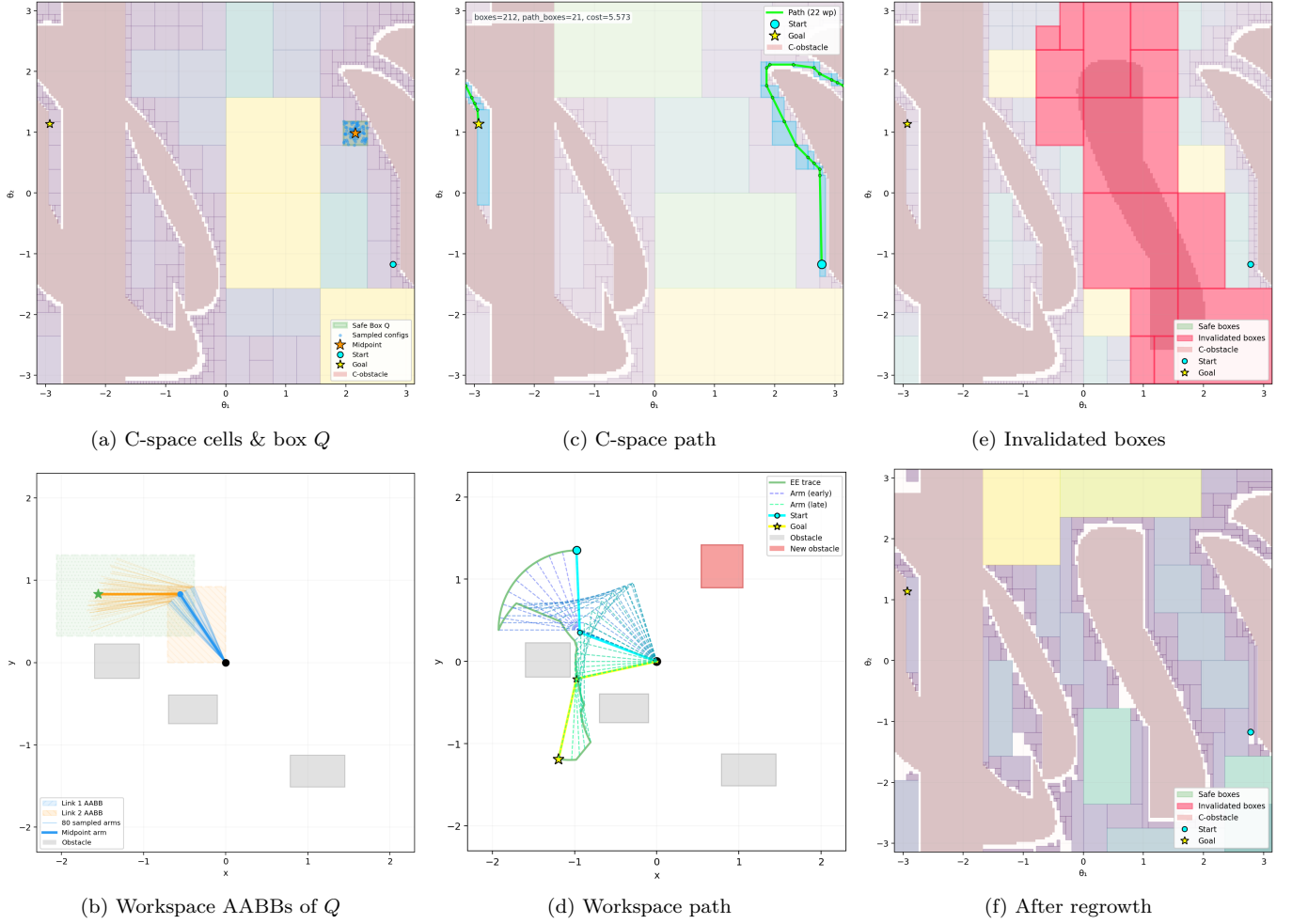


Fig. 2. Illustrative RBF build, query, and obstacle-update cycle on a planar 2-link manipulator. (a, b) Adaptive C -space cells and workspace AABB envelopes of the highlighted box Q . (c, d) Box-corridor path in C -space and its workspace trajectory. (e, f) After obstacle insertion, intersecting boxes are invalidated (red), and localized repair/regrowth updates the explicit forest without requiring a full rebuild.

possible under the available box, time, and depth budgets, and the induced adjacency graph supports efficient downstream search. This object is scene-specific: changing $\mathcal{O}$ changes which intervals are accepted into the forest, even though the underlying envelope evidence in LECT remains reusable.

The central structural rule is that the forest is not merely a free-space cover, but a *corridor-compatible* free-space cover. Boxes are inserted under a connected-box constraint.

Viewed as adaptive cell decomposition, this objective is intentionally workload-biased rather than globally exhaustive. The planner does not try to classify every cell in Q . It refines and accepts cells near useful frontiers, task seeds, and connector candidates, then keeps the resulting adjacency graph as the reusable planning object.

a) *Connected-box rule.*: For boxes

$$B_i = \prod_{d=1}^n [l_i^{(d)}, u_i^{(d)}], \quad B_j = \prod_{d=1}^n [l_j^{(d)}, u_j^{(d)}],$$

RBF treats them as connected when, in every coordinate, the two intervals either overlap or are separated by at most a small tolerance ε_c :

$$\max\{l_i^{(d)}, l_j^{(d)}\} \leq \min\{u_i^{(d)}, u_j^{(d)}\} + \varepsilon_c, \quad d = 1, \dots, n.$$

When $\varepsilon_c = 0$, this reduces to ordinary interval intersection or boundary touching in every dimension. A small positive tolerance absorbs the boundary offset used by grower seeds and floating-point roundoff.

This rule is deliberately weaker than requiring a full shared face. It treats overlap, touching, and tolerance-scale gaps uniformly as graph-connectivity evidence, while validity remains attached to the accepted box union itself. The output of the construction stage is a forest of validated boxes whose graph connectivity is aligned with the corridor object queried later, without imposing a stricter face-sharing invariant than the grower actually maintains.

B. Find Free Box

FINDFREEBOX descends LECT from the root toward the leaf containing seed $q \in Q$, materializing endpoint and link envelopes lazily along the path, skipping saturated subtrees via occupancy metadata, and returning the current interval as a validated box candidate as soon as the scene oracle accepts it. Tree topology and endpoint records are reused from the scene-independent cache, while scene-specific validation is deferred to each grower visit. Algorithm 1 gives the complete pseudocode.

Algorithm 1 FindFreeBox: local box materialization around a seed.

Input: Seed configuration q , LECT T , obstacle set $\mathcal{O}$, depth cap $D_{\max}$

Output: Validated interval I_v or failure

```

1:  $v \leftarrow \text{root}(T)$ ;  $d \leftarrow 0$ ; initialize current interval and FK state
2: while  $d \leq D_{\max}$  do
3:   Cache materialization
4:   materialize the endpoint and link envelopes of  $v$  if needed
5:   Scene validation
6:   if  $v$  is marked fully occupied then
7:     return failure
8:   end if
9:   if  $v$  is validated collision-free with respect to  $\mathcal{O}$  then
10:    return interval  $I_v$ 
11:   end if
12:   Refinement
13:   if  $d = D_{\max}$  then
14:     return failure
15:   end if
16:   if  $v$  has no children then
17:     split  $v$  along the depth-synchronous split dimension of
       depth  $d$ 
18:   end if
19:    $k \leftarrow$  split dimension of  $v$ 
20:    $v \leftarrow$  the child of  $v$  whose interval contains  $q$ ; update the
     current interval
21:   update the FK state from dimension  $k$  onward;  $d \leftarrow d + 1$ 
22: end while
23: return failure

```

The termination behavior is straightforward. Under depth cap $D_{\max}$, each loop iteration either returns immediately or descends one more level in the tree, so the procedure can make at most $D_{\max} + 1$ decisions. In strict conservative mode, any returned interval has passed envelope disjointness against $\mathcal{O}$; in performance-oriented modes, the returned interval is a validated planning region that remains subject to the common final audit used for reported paths.

C. Forest Growth

Starting from an initial seed set, the forest is grown by repeatedly invoking the find-free-box procedure on carefully chosen candidate configurations. Each seed first produces a root box through the find-free-box procedure, and these root boxes initialize one or more connected components of the forest. The main grower should be understood as an evidence-guided rapidly exploring process rather than as a generic random cover: every iteration chooses a frontier component, proposes a nearby interval worth testing, and accepts the result only when it extends the forest by a usable connected corridor segment.

Each iteration makes two explicit heuristic decisions: it samples a target configuration q_t , then chooses a boundary seed on an existing box face that is closest to that target. The reported planning configuration uses a sequential Bernoulli schedule with component-connector probability $p_c = 0.45$, query-root bias $p_g = 0.20$, intertree-root bias $p_i = 0.25$, and unexplored-volume probability $p_u = 0.45$. A component-connector target between disconnected root components is attempted first. If no connector target is used, coordinated multi-tree mode may target a root from another active component; because $p_i \leq 0.5$ in the reported runs, no high-bias pulse is active and the sustained

cap is also 0.25. If no intertree target is used, the grower then tries a query-root target, then a LECT unexplored-volume leaf, and finally a uniform root-interval sample. The same reported configuration uses 5000 boxes, a 60 s build timeout, an FFB depth cap of 120, eight worker threads, a component-connector candidate limit of four, a staged component-connect radius of 0.35 in normalized L_∞ distance, a post-connect time budget of 450 ms, and a quality floor of 64 connected boxes. These numbers are planner settings rather than envelope-theory parameters, but they are part of the self-contained specification for the reported rows.

Given a target q_t and an existing box $B = \prod_j [\ell_j, u_j]$, the grower enumerates feasible faces $f = (B, d, s)$, where d is a joint dimension and $s \in \{-1, +1\}$ is the lower or upper side. Let τ be the adjacency tolerance and $\epsilon_b = \max(\epsilon_{\text{guard}}, \tau/4)$. A face is feasible only if stepping outside it by ϵ_b remains inside the global root interval. In the reported configurations $\epsilon_{\text{guard}} = 0$, so this formula uses the $\tau/4$ floor. The seed associated with the face is

$$q_f[j] = \begin{cases} u_d + \epsilon_b, & j = d, s = +1, \\ \ell_d - \epsilon_b, & j = d, s = -1, \\ \text{clip}(q_t[j], \ell_j + \epsilon_b, u_j - \epsilon_b), & j \neq d, \end{cases} \quad (1)$$

with clipping to $[\ell_j, u_j]$ when a box is thinner than $2\epsilon_b$. Faces are ranked by the squared target distance

$$\sigma(f; q_t) = \|q_t - q_f\|_2^2, \quad (2)$$

and the implementation retains the 128 lowest-score face candidates across the scanned frontier boxes. It then tries them in increasing score order, skipping a candidate if the corresponding LECT leaf is temporarily cooled or if a component-connection cache has already recorded the same boundary seed. The first remaining candidate becomes the seed for FINDFREEBOX.

The candidate box is accepted only if FINDFREEBOX validates it and the new box is connected to its parent under the same adjacency predicate used by the query graph: all dimensions overlap up to tolerance τ , and at least one dimension touches within tolerance or all dimensions overlap. Failures are also handled explicitly. When the LECT leaf containing a failed seed reaches the hard-frontier failure threshold of one failed attempt, the leaf is skipped for a horizon of 300 subsequently accepted boxes, unless a later depth stage raises the active search depth and enables a retry. Cooling and the frontier-seed cache do not alter the validation boundary; they only reduce repeated calls on already failed or already covered frontier regions. The resulting method is summarized in algorithm 2.

The main reported configuration uses coordinated multi-goal growth. Five seed configurations initialize the shelf roots, and the grower uses the fixed resource budget defined by the planner configuration. This choice keeps the growth schedule simple enough to preserve one authoritative forest state while still exposing reusable LECT evidence across threads and cache-warm reruns.

Algorithm 2 GrowForest: budgeted expansion of a connected box-region forest.

Input: LECT T , obstacle set $\mathcal{O}$, seed set $\mathcal{S}$, box budget $N_{\max}$, time budget $t_{\max}$
Output: Validated box-region forest $\mathcal{F}$

```

1:  $\mathcal{F} \leftarrow \emptyset$ ; initialize connected components from  $\mathcal{S}$ 
2: for all  $q \in \mathcal{S}$  do
3:    $B \leftarrow \text{FindFreeBox}(q, T, \mathcal{O})$ 
4:   if  $B \neq \emptyset$  then
5:     insert  $B$  into  $\mathcal{F}$  as a root box
6:   end if
7: end for
8: while  $|\mathcal{F}| < N_{\max}$  and time budget not exhausted do
9:   draw target  $q_t$ : connector with prob.  $p_c$ , query/intertree root with prob.  $p_g$  or  $p_i$ , unexplored LECT leaf with prob.  $p_u$ , otherwise uniform
10:  enumerate feasible faces  $f = (B, d, s)$  of frontier boxes and compute  $q_f$  and  $\sigma(f; q_t)$ 
11:  choose the first of the 128 lowest-score face seeds that is not cooled and not in the frontier-seed cache
12:  set  $(q_{\text{seed}}, B_{\text{near}}) \leftarrow (q_f, B)$  for that face
13:   $B_{\text{new}} \leftarrow \text{FindFreeBox}(q_{\text{seed}}, T, \mathcal{O})$ 
14:  if  $B_{\text{new}} \neq \emptyset$  and  $\text{BoxesConnected}(B_{\text{new}}, B_{\text{near}}, \tau)$  then
15:    insert  $B_{\text{new}}$  into  $\mathcal{F}$ ;  $\text{UpdateComponents}(\mathcal{F})$ 
16:  else
17:    record the failed LECT leaf; cool it after one failed attempt for a 300-box horizon
18:  end if
19: end while
20: return  $\mathcal{F}$ 

```

The regenerated quantitative section in this version focuses on the endpoint envelope primitive, so this planning configuration is described as the implementation surface rather than as a reported timing row. The output of this section is the query-ready box-region object only.

D. Forest Consolidation

Raw growth tends to produce a forest that is finer than necessary. The purpose of consolidation is to replace a large collection of locally valid boxes by a smaller collection of equally valid but more useful regions. Two operations are central.

The first is *promotion*. When two occupied sibling leaves are both collision-free and their parent interval is also collision-free, the two leaves can be replaced by the parent. This is the most local form of coarsening and directly reuses the multiresolution structure of LECT.

The second is *multi-level coarsening*. Exact connected merges are the cleanest case: if two boxes are connected and their union is itself one box, the merged region can be accepted directly after validation. More general connected candidate merges are handled by constructing a candidate parent region and accepting it only when its chosen envelope representation remains collision-free against the current obstacle set. This is where the representation choices of Sections III and IV become operational. In an AABB-based forest, the compact parent witness is typically the enclosing AABB of the two child boxes. In a SupportHull-based forest, parent witnesses are kept compact only when the merged endpoint-hull witness remains acceptable; otherwise the child witnesses are kept. Greedy non-sibling candidates are ranked by a volume-expansion score, so exact face merges are preferred and high-inflation parents are deferred unless they pass validation under the

Algorithm 3 ConsolidateForest: graph-preserving promotion, merging, and pruning.

Input: Validated forest $\mathcal{F}$, LECT T , obstacle set $\mathcal{O}$
Output: Consolidated validated forest $\mathcal{F}'$

```

1:  $\mathcal{F}' \leftarrow \mathcal{F}$ 
2: Stage 1: sibling promotion
3: repeat
4:   promote any validated sibling leaves whose parent is collision-free
5: until no eligible sibling pair remains
6: Stage 2: validated connected/contact merging
7: merge exact connected pairs whose union is itself a single box
8: for all connected candidate pairs  $(B_i, B_j)$  in  $\mathcal{F}'$  do
9:   build a candidate parent region  $H$  and score its volume expansion
10:  if the chosen envelope representation of  $H$  is collision-free with respect to  $\mathcal{O}$  then
11:    replace  $B_i, B_j$  by  $H$ 
12:  end if
13: end for
14: Stage 3: containment pruning
15: remove only contained boxes whose adjacency and segment witnesses transfer to selected covering regions
16: return  $\mathcal{F}'$ 

```

scene oracle. The consolidation logic is summarized in Algorithm 3. The final containment-pruning step removes only boxes that are fully covered by accepted validated regions and whose graph incidence can be transferred to a selected covering region. It changes neither the represented free-space union nor the corridor witnesses used by query search.

Validity follows directly: every accepted promotion or merge has itself passed the scene oracle, so the represented free-space union cannot shrink. Containment pruning removes only boxes fully covered by a selected region that inherits their adjacency witnesses, preserving all corridor connectivity.

E. Corridor Query

After growth and consolidation, the forest is exported as a graph $G = (V, E)$ with one vertex per validated box. Two vertices are connected by an edge when their boxes satisfy the connected-box rule above. Lightweight querying is possible because this connected-box structure is enforced during growth and consolidation rather than repaired after graph export.

If disconnected adjacency islands remain after growth, an optional connector stage attempts targeted bidirectional sampling bridges between nearest frontier boxes of different components. A successful bridge is stored as a segment edge rather than as a claim that the two boxes overlap. This distinction is important for both the RBF query layer and the GCS negative controls: a segment edge is a collision-checked path witness between two boxes, whereas a GCS edge over convex regions requires feasible overlap of the regions themselves.

This pair $(\mathcal{F}, G)$ is the final planning object produced by the build stage. Querying it is intentionally lightweight. Given start and goal configurations q_s, q_g , the planner associates them with start-side and goal-side boxes in $\mathcal{F}$ and then searches G for a connecting box sequence. The

query problem is reduced to box-corridor retrieval inside an already constructed free-space representation.

A shortest path on G yields a box sequence whose union defines a corridor between the two query endpoints. A geometric path is then recovered as a waypoint chain inside that corridor. In the experiments, this candidate is then passed through the same OMPL-style simplification/local post-processing and fixed-step final audit used by the comparison pipelines. That post-processing can move the final reported path outside the source box union. Consequently, the box-corridor statement below applies only to contained paths before or after post-processing, while the reported planning success in the main curves is the common fixed-resolution audit outcome.

Theorem 1 (Conditional validated box-corridor path). *Let $\mathcal{F}$ be a forest whose boxes have each passed conservative link-envelope validation against obstacle set $\mathcal{O}$. If a continuous path $\gamma : [0, 1] \rightarrow \mathcal{Q}$ satisfies $\gamma([0, 1]) \subseteq \bigcup_{B \in \mathcal{F}} B$, then every configuration on γ is collision-free with respect to $\mathcal{O}$.*

Proof. For any $s \in [0, 1]$, the containment assumption gives a validated box $B \in \mathcal{F}$ with $\gamma(s) \in B$. By construction, validation of B means every conservative link envelope for B is disjoint from $\mathcal{O}$. Because each such envelope contains the link geometry for every configuration inside B (section III), the robot geometry at $\gamma(s)$ is also disjoint from $\mathcal{O}$. Since s was arbitrary, the whole path is collision-free. $\square$

a) Reporting protocol.: Theorem 1 is a conditional statement: it covers path portions contained in a conservatively validated box union. It is not the acceptance rule for the main experimental comparisons. When OMPL simplification, local repair, smoothing, or external composition leaves that union, the path is accepted only if it passes the same fixed-resolution final scene collision audit used for the baselines. The main performance curves should therefore be read as post-processed, fixed-audit planning results rather than as evidence that every reported path satisfies the box-corridor containment hypothesis.

For dynamic scenes, the same explicit forest representation supports local graph maintenance without compulsory rebuilds. Each accepted box records the obstacle evidence that blocked or validated its envelope tests; obstacle deletions cannot invalidate existing boxes accepted under the same validation policy, because free space only grows. Deletion triggers only a spatial dirty-region pass around the removed obstacle to choose local regrowth seeds, after which any new box is validated by the same oracle as in the original build. Obstacle insertion is stricter: the new obstacle is inflated by the dirty-region padding, all overlapping boxes are rechecked exactly, invalid boxes and incident graph edges are removed, and every surviving segment edge is re-audited in the edited scene. The repair stage then regrows from deleted-box centers, live neighbours, original task seeds, and component-boundary candidates; each committed box must pass the active validation policy before it enters the forest. Finally, query failure can trigger

a bounded local repair search, again filtered by the same final audit. Warm rebuild remains available as a measured baseline and fallback policy.

If no contained conservative corridor is recovered, no box-corridor guarantee attaches to that query. The construction still exports a concrete box-region planning object, and section VI evaluates its build cost, query cost, path quality, and success rate after the same post-processing and fixed-resolution final audit used for the other planners.

VI. Experiments

This section reports the regenerated endpoint-envelope source study (Experiment 1) and the width-wise link-envelope representation study (Experiment 2). Together they measure the geometric primitives used by the later box-validation and planner layers, not planner-level path benchmarks.

a) Protocol.: The benchmark uses the IIWA14 model and six fixed joint-box widths $\{0.02, 0.05, 0.10, 0.20, 0.30, 0.50\}$ rad. For each width, 1000 shared joint boxes are generated from the same seeded table and then replayed by all endpoint sources, giving 6000 endpoint boxes per source. The evaluated sources are IFK (the public spelling that now subsumes the former AAFK alias), HIFK with depths 3 and 5, CritSample, Analytical, and MC. MC uses width-proportional sample counts from 2857 to 71429 samples per box, with 50000 reference samples at the nominal width. The output artifact stores the fixed box table hash so the link-envelope study can reuse the same corpus.

b) Metrics.: For each source and width, the runner records endpoint construction time, endpoint AABB volume, implementation certification status, and the diagnostic gap to the finite reference union of CritSample, Analytical, and MC records. Table I restores the original Table I structure and reports the mean endpoint AABB volume V , mean evaluation time, and maximum observed gap at each fixed width. The gap column is a diagnostic over the sampled corpus, not a formal containment proof.

A. Endpoint Envelope Source Study

Table I summarizes the endpoint interval-envelope source comparison used to choose the geometric inputs for the current paper track. The IFK row now stands for the public spelling whose implementation is identical to the old AAFK alias, so it is reported only once. IFK remains the cheapest certified source and stays gap-free against the sampled reference union over all reported widths, with mean endpoint time between 4.3 and 4.6 μ s.

Hierarchical IFK trades time for tighter certified boxes. All three certified AA rows remain gap-free against the sampled reference union over the reported widths. After fixing HIFK's split-policy selection and rerunning the same 6000 shared boxes, HIFK no longer pays the previous online dimension-search overhead. At width 0.50 rad, the mean endpoint volume drops from 5.746 m^3 for IFK to

TABLE I
Endpoint-interval AABB source comparison at fixed widths.

Width	Source	V (m ³)	Mean (μ s)	Max gap (m)
0.02	IFK	0.000049	4.3	0.0000
0.02	HIFK_3	0.000047	23.9	0.0000
0.02	HIFK_5	0.000047	73.1	0.0000
0.02	CritSample	0.000046	10.0	0.0000
0.02	Analytical	0.000046	6475.9	0.0000
0.02	MC	0.000029	2263.1	0.0061
0.05	IFK	0.000872	4.5	0.0000
0.05	HIFK_3	0.000784	24.1	0.0000
0.05	HIFK_5	0.000757	72.5	0.0000
0.05	CritSample	0.000712	10.7	0.0003
0.05	Analytical	0.000713	6609.5	0.0000
0.05	MC	0.000499	5603.3	0.0145
0.10	IFK	0.008530	4.6	0.0000
0.10	HIFK_3	0.006907	24.1	0.0000
0.10	HIFK_5	0.006432	72.0	0.0000
0.10	CritSample	0.005686	11.1	0.0012
0.10	Analytical	0.005690	6753.7	0.0000
0.10	MC	0.004201	11412.3	0.0307
0.20	IFK	0.1023	4.6	0.0000
0.20	HIFK_3	0.0676	26.4	0.0000
0.20	HIFK_5	0.0583	80.2	0.0000
0.20	CritSample	0.0451	12.7	0.0054
0.20	Analytical	0.0452	7074.8	0.0000
0.20	MC	0.0351	22866.2	0.0702
0.30	IFK	0.5222	4.6	0.0000
0.30	HIFK_3	0.2819	23.4	0.0000
0.30	HIFK_5	0.2246	70.8	0.0000
0.30	CritSample	0.1495	13.9	0.0126
0.30	Analytical	0.1505	7218.2	0.0000
0.30	MC	0.1202	34588.3	0.0956
0.50	IFK	5.746	4.6	0.0000
0.50	HIFK_3	2.073	24.1	0.0000
0.50	HIFK_5	1.381	72.2	0.0000
0.50	CritSample	0.6559	18.0	0.0324
0.50	Analytical	0.6657	7489.0	0.0030
0.50	MC	0.5476	57728.9	0.1137

2.073 m³ for HIFK_3 and 1.381 m³ for HIFK_5, while mean endpoint cost rises from 4.6 to 24.1 to 72.2 μ s. This preserves the same qualitative trade-off as the old Table I, but now places HIFK in a practical middle tier: HIFK_3 removes most of IFK’s wide-box slack, while HIFK_5 pushes that trade-off further without the earlier artificial timing plateau.

The non-certified sources are useful as tightness references but are not safe drop-in replacements for certified validation. CritSample still tracks Analytical much more closely than any certified AA source on wide boxes, reaching 0.6559 versus 0.6657 m³ at 0.50 rad while remaining orders of magnitude cheaper, 18.0 versus 7489.0 μ s, but it exhibits a 0.0324 m maximum observed gap against the finite reference union. MC is the tightest row at each reported width, but it is also the slowest and shows the largest observed gap, reaching 0.1137 m at 0.50 rad. The regenerated Exp.1 result therefore supports the paper’s default design choice: use IFK for cheap certified envelopes, and reserve hierarchical or sampling-based sources for tighter diagnostics and future ablation studies.

B. Link-Envelope Representation Study

Table II restores the original mainline representation study with the certified IFK_AA endpoint source and the fixed short-link split $S = 1$. The benchmark reuses the same six fixed widths and shared box corpus as Experiment 1 and compares only the retained mainline

TABLE II
Width-wise IFK_AA link-envelope comparison.

Width	Envelope	V (m ³)	Build (μ s)	Collision (μ s)
0.02	LinkIAABB	0.0136	0.1	0.5
0.02	SupportHull	0.000379	0.2	0.8
0.05	LinkIAABB	0.0199	0.1	0.5
0.05	SupportHull	0.0028	0.2	0.9
0.10	LinkIAABB	0.0393	0.1	0.5
0.10	SupportHull	0.0151	0.2	1.1
0.20	LinkIAABB	0.1508	0.1	0.6
0.20	SupportHull	0.1101	0.2	1.7
0.30	LinkIAABB	0.5193	0.1	0.8
0.30	SupportHull	0.4602	0.2	2.7
0.50	LinkIAABB	4.691	0.1	1.1
0.50	SupportHull	4.617	0.2	6.7

representations, LinkIAABB and pure SupportHull, on uninflated short-link record volume, envelope construction time, and envelope-only obstacle-check time.

The corrected volume proxy tightens monotonically from LinkIAABB to pure SupportHull at every width. The gap is most visible on narrow boxes: at 0.02 rad the reported volume drops from 0.0136 m³ for LinkIAABB to 0.000379 m³ for SupportHull. As the width grows, the two rows converge, but the ordering remains unchanged even at 0.50 rad (4.691 versus 4.617 m³). LinkIAABB remains the cheapest row to construct, staying near 0.1 μ s across the sweep, while pure SupportHull remains compact at about 0.2 μ s once the `keep_kdop=false` path stops retaining an extra KDOP payload slot. The obstacle-check time shows the expected mainline trade-off: LinkIAABB stays cheapest, rising from about 0.5 to 1.1 μ s, while pure SupportHull rises from about 0.8 to 6.7 μ s as the boxes widen and more survivors reach the GJK refinement stage. The retained mainline representation set therefore reduces to a simple coarse-versus-tight trade-off: AABB for the cheapest broad filter, and pure SupportHull when the tighter directional witness is worth the extra runtime.

C. LECT Disk Microbenchmark

Experiment 3 reports the real endpoint-only Shelf+IIWA p18 persisted cache used by the warm-build study and the immutable read snapshot used by the planner read path. The cache is built once by an 8-thread p18 prewarm that materializes only the depth-18 leaf endpoint envelopes and then lets the database propagate child hulls upward through the remaining 17 levels. The rebuilt mutable namespace stores 524,287 nodes and 524,287 endpoint-envelope records, occupies 443.07 MB on disk, and touches 262,144 depth-18 leaves during the one-time prewarm. Publishing the immutable read snapshot adds a 405.45 MB mmap-only artifact. Table III reports the prewarm and snapshot publication costs together with the planner read-lane microbenchmark on that snapshot. Strict verification remains available as an offline audit, but it is no longer folded into the timed open row because it measures full consistency checking rather than planner read-path latency.

TABLE III

LECT endpoint-only persisted-cache microbenchmark on the real Shelf+IIWA p18 cache. The prewarm and snapshot-build rows are one-time publication steps; the remaining rows open and query the immutable read snapshot, reporting average per operation using the most readable unit for each row.

Operation	Ops	Avg.
P18 prewarm (8 threads)	1	72.365 s
Snapshot build	1	242.606 ms
Snapshot open read-only	1	0.026 ms
Node box	20000	0.473 μ s
Exact box lookup	20000	0.409 μ s
Exact endpoint lookup	20000	0.707 μ s
Range query	20000	41.473 μ s
Evidence read	20000	0.190 μ s

Table III shows that the endpoint-only persisted cache keeps the planner read path lightweight after the one-time p18 prewarm and snapshot publication. On the real 524k-node Shelf+IIWA artifact, opening the immutable read snapshot costs 0.026 ms, point evidence read is 0.190 μ s, exact box lookup is 0.409 μ s, and the fused exact-endpoint lookup used by the warm external-evidence path is 0.707 μ s. The wider intersecting range-query stage remains tens of microseconds at 41.473 μ s, still far below rebuilding the cache. The heavier offline steps are the one-time p18 prewarm at 72.365 s and snapshot publication at 242.606 ms; both are paid once per published cache generation rather than once per planner open.

VII. Conclusion

This paper presented RBF, a link-envelope-guided box-forest method for fast multi-query manipulator planning. Its geometric core maps endpoint interval envelopes to link interval envelopes via capsule Minkowski decomposition and convex-hull generator bounding; an RRT-inspired grower builds a scene-specific box forest into a corridor graph queried by graph search and local refinement; LECT optionally caches kinematic envelope records to reduce repeated materialization cost. The regenerated experiments in this version evaluate the endpoint-envelope primitive directly and add a certified IFK-based $S = 1$ link-envelope representation study plus a disk-backed LECT microbenchmark: IFK is the fastest certified endpoint row, LinkIAABB is the lightest retained link witness, and the real Shelf+IIWA persisted-cache study shows that evidence reads stay microsecond-scale, exact box lookup stays sub-millisecond, and whole-store checkpoint/compact remain second-scale maintenance operations on the full 524k-node cache. Older planner-level cache, dynamic-update, and cross-planner tables are intentionally not carried forward.

a) Limitations.: The box-corridor guarantee (theorem 1) is conditional on conservative IFK validation and path containment inside the validated box union; the regenerated Exp.1 and Exp.2 rows do not exercise full planner paths and therefore do not claim that condition. The Exp.1 gap diagnostic is computed against a finite union of CritSample, Analytical, and MC records, so it

should be read as an implementation sanity check rather than as a formal proof of containment.

b) Scope and Future Work.: The empirical scope in this version covers the endpoint-envelope source sweep, a width-wise IFK/ $S = 1$ link-envelope representation study, and a real Shelf+IIWA persisted-cache microbenchmark. The next steps are to extend the LECT study from storage-level timings to Shelf+IIWA warm-build rows, planner-level cache reuse, dynamic-update rows, and broader cross-planner or multi-robot comparisons under the same fixed-artifact protocol.

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